

PROGRAM:

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#include<stdio.h>
#define MAX_SIZE 100
void createArray(int arr[], int *size);
void insertElement(int arr[], int *size, int element, int position);
int searchElement(int arr[], int size, int element);
void deleteElement(int arr[], int *size, int position);
void displayArray(int arr[], int size);
int main() {
    int arr[MAX_SIZE];
    int size = 0;
    int choice, element, position, result;
    while(1) {
        printf("\nArray Operation Menu:\n");
        printf("1. Create Array\n");
        printf("2. Insert Element\n");
        printf("3. Search Element\n");
        printf("4. Delete Element\n");
        printf("5. Display Array\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                createArray(arr, &size);
                break;
            case 2:
                printf("Enter element to insert: ");
                scanf("%d", &element);
                printf("Enter position to insert (0 to %d): ", size);
                scanf("%d", &position);
                insertElement(arr, &size, element, position);
                break;
            case 3:
                printf("Enter element to search: ");
                scanf("%d", &element);
                result = searchElement(arr, size, element);
                if(result != -1)
                    printf("Element found at position: %d\n", result);
                else
                    printf("Element not found in the array\n");
                break;
            case 4:
                printf("Enter position to delete (0 to %d): ", size-1);
                scanf("%d", &position);
                deleteElement(arr, &size, position);
                break;
            case 5:
                displayArray(arr, size);
                break;
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        case 6:
            return 0;
        default:
            printf("Invalid choice! Please enter a valid option.\n");
    }
}
return 0;
}

void createArray(int arr[], int *size) {
    int n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    if(n > MAX_SIZE) {
        printf("Error: Number of elements exceeds the maximum array size.\n");
        return;
    }
    printf("Enter %d elements: ", n);
    for(i=0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    *size = n;
    printf("Array created successfully.\n");
}

void insertElement(int arr[], int *size, int element, int position) {
    int i;
    if(*size >= MAX_SIZE) {
        printf("Error: Array is full. Cannot insert element.\n");
        return;
    }
    if((position < 0) || (position > *size)) {
        printf("Error: Invalid position.\n");
        return;
    }
    for(i=0; i < *size; i++) {
        arr[i] = arr[i - 1];
    }
    arr[position] = element;
    (*size)++;
    printf("Element inserted successfully.\n");
}

int searchElement(int arr[], int *size, int position) {
    int i;
    for(i = 0; i < *size; i++) {
        if(arr[i] == element) {
            return i;
        }
    }
    return -1;
}

void deleteElement(int arr[], int *size, int position) {
    int i;

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        if((position < 0) || (position >= *size)) {
            printf("Error: Invalid position.\n");
            return;
        }
        for(i = position; i < *size - 1; i++) {
            arr[i] = arr[i + 1];
        }
        (*size) --;
        printf("Element deleted successfully.\n");
    }
    void displayArray(int arr[], int size) {
        int i;
        if(size == 0) {
            printf("Array is empty.\n");
            return;
        }
        printf("Array elements: ");
        for(i = 0; i < size; i++) {
            printf("%d", arr[i]);
        }
        printf("\n");
    }
}

```

OUTPUT:

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 1

Enter the number of elements: 5

Enter 5 elements: 1 2 3 4 5

Array created successfully.

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 2

Enter element to insert: 10

Enter position to insert (0 to 5): 2

Element inserted successfully.

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element

4. Delete Element

5. Display Array

6. Exit

Enter your choice: 5

Array elements: 1 2 10 3 4 5

Array Operations Menu:

1. Create Array

2. Insert Element

3. Search Element

4. Delete Element

5. Display Array

6. Exit

Enter your choice: 3

Enter the element to search: 10

Element found at position: 2

Array Operations Menu:

1. Create Array

2. Insert Element

3. Search Element

4. Delete Element

5. Display Array

6. Exit

Enter your choice: 4

Enter position to delete (0 to 5): 2

Element deleted successfully.

Array Operations Menu:

1. Create Array

2. Insert Element

3. Search Element

4. Delete Element

5. Display Array

6. Exit

Enter your choice: 5

Array elements: 1 2 3 4 5